

# Supervised ML Workflow

Or Hirshman Shavit

# Agenda

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We will walk through the End-to-End Supervised ML Workflow

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Workflow

From Raw Data to Production Models.

Define Objective

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A vague goal often leads to a failed model.

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**Objectives:**

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Define the real-world problem you are solving  
(e.g., "Reduce customer churn by 15%").

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Translating the business goal into a mathematical task

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Translating the business goal into a mathematical task

(e.g., "Binary classification to predict probability of churn")

# Defining the Model Objective

**Key Questions to Answer:**

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How will the prediction be used? will it trigger an automated action?

What is the baseline? (What is the performance of the current manual process or simple heuristic?)

# Defining the Model Objective

## **Key Questions to Answer:**

What is the desired output? (A category? numerical value? ranking?)

How will the prediction be used? will it trigger an automated action?

What is the baseline? (What is the performance of the current manual process or simple heuristic?)

Success Metrics (KPIs): Define quantifiable targets (e.g., "F1-Score > 0.85").

# Defining the Model Objective

Clearly define the business question and target variable

Collect Data

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## **Diverse Sourcing Strategies:**

- Internal Data:

Organization databases, log files.

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## **Diverse Sourcing Strategies:**

- Internal Data:

Organization databases, log files.

- External Data:

Public repositories, Kaggle, scrapers, Data purchased from partners, Google Dataset Search, etc...

# Collect Data

**Note:**

Take care about duplicated records and merge results.

EDA

# EDA

**The idea:**

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Before modeling, we must know our data.

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EDA is an iterative process of visualizing and summarizing to find patterns.

# EDA

## **Multivariate Analysis:**

Using scatter plots, pair plots and heatmaps to find correlations between features and target variable.

# EDA

## **Statistics Methods:**

We can use statistics methods such as t-test, anova, chi-square to find relations between features and target variable.

EDA

**Outcomes:**

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Does the data actually reflect the real-world business case?

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2. Now we can validate assumptions-

Does the data actually reflect the real-world business case?

3. We now have more ideas about what are the important features, which features to engineer, which outliers to cap, and which columns to drop entirely.

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2. CV (Train (80%), Test (20%).
3. Stratification: Mandatory for classification with imbalance target.
4. Time split - when time is crucial

# Handling Outliers

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**Detection:**

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## **Detection:**

We can use:

1. Visualizations
2. Z-score
3. IQR
4. Isolation Forest

# Handling Outliers

**Treatment:**

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- Trimming/Removal: Only if outliers are confirmed errors or entry mistakes.

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- Data Transformation

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## **Treatment:**

- Trimming/Removal: Only if outliers are confirmed errors or entry mistakes.
- Capping: Clamping values at 1st and 99th percentiles.
- Data Transformation
- Use ML models that handles them inheritably.

# Handling Missing Values

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## **Detection:**

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## **Detection:**

Identify what types of missing data we have

(Random, Fully Random and Not Random)

# Handling Missing Values

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# Handling Missing Values

## **Treatment:**

- Simple Imputation: Mean/Median for numerical, Mode for categorical.
- Interpolation
- Model-Based Imputation: MICE or KNN-Imputer.
- Flagging: Preserve the signal of why data was missing.

# Categorical Data Handling

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for nominal - no order and low cardinality.

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Explicit mapping (Small=1, Med=2, Large=3)

to preserve hierarchy.

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for nominal - no order and low cardinality.

## 2. Ordinal (Ranked Order):

Explicit mapping (Small=1, Med=2, Large=3)

to preserve hierarchy.

## 3. Target Encoding:

For high cardinality. Avoid data leakage.

# Feature Scaling and Transformation

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Squeezing data into  $[0, 1]$ .

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3. Log transformations:

Force skewed distributions into a "Gaussian-like" bell curve.

# Imbalanced Data

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In case:

1. We do classification

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3. Train a baseline model
4. Check multiple imbalanced techniques:
  - a. ROS
  - b. RUS
  - c. SMOTE
  - d. SMOTE ENN

# Imbalanced Data

Now compare performance to baseline model.

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Always choose method according to data size limitations.

# Model Selection

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2. Check multiple metrics - always follow the business objectives!
3. Simplicity
4. Running time

# Hyperparameter Tuning

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Run on top of the chosen model

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Tries every hyperparameter combination.

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- Random Search:

# Hyperparameter Tuning

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- Grid Search:

Tries every hyperparameter combination.

- Random Search:

Superior to for high-dimensional spaces.

# Hyperparameter Tuning

**Notes:**

# Hyperparameter Tuning

## **Notes:**

1. Conducted solely on the Validation Set.

# Hyperparameter Tuning

## **Notes:**

1. Conducted solely on the Validation Set.
2. Choose simplest hyperparameters with good performance to avoid overfitting.

# Final Evaluation

# Final Evaluation & Deployment

The Moment of Truth:

Once the model is tuned, evaluate it exactly once on the Test Set.

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Success Metrics:

Following business perspective.

# Documentation

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1. Detailed report of the whole ML process.

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Data scientist create 2 documentations:

1. Detailed report of the whole ML process.
2. Short presentation for stakeholders.

Documentation

**Model Interpretability & Bias Detection**

# Documentation

## **Model Interpretability & Bias Detection**

Even a high-accuracy model can be rejected if it is a "black box" or contains hidden biases.

# Documentation

Before deployment, a few critical step are added to the workflow:

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1. Explainability: Using techniques like SHAP to understand which features drove a specific prediction.

# Documentation

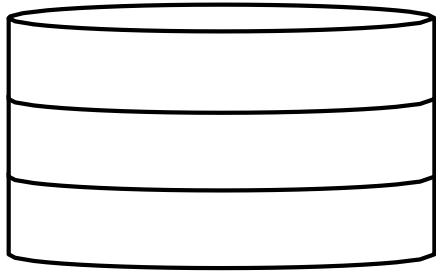
Before deployment, a few critical step are added to the workflow:

1. Explainability: Using techniques like SHAP to understand which features drove a specific prediction.
2. Responsible AI & Bias Auditing: Checking for Disparate Impact across demographics (e.g., age, gender) to ensure the model doesn't unfairly penalize certain groups.

# Deployment

# ML Architecture

Historical Data



Video/Audio

Structured

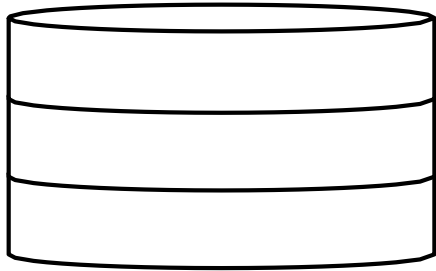
Free text

files

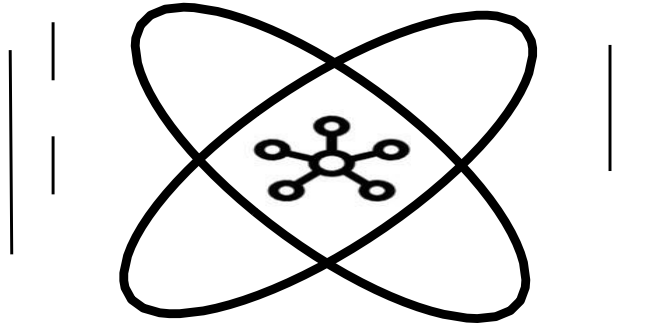
tables

# ML Architecture

Historical Data



ML Model



Video/Audio

Structured

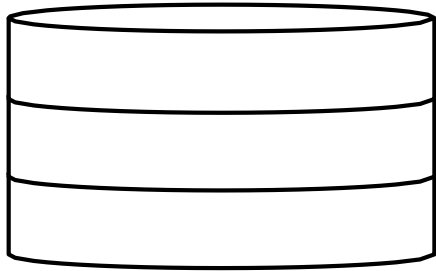
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# ML Architecture

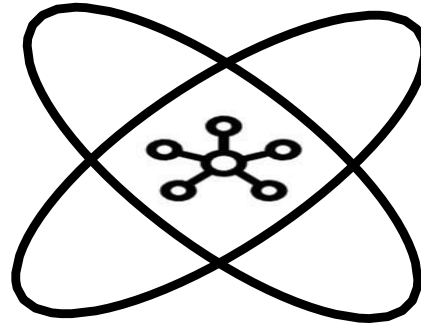
Historical Data



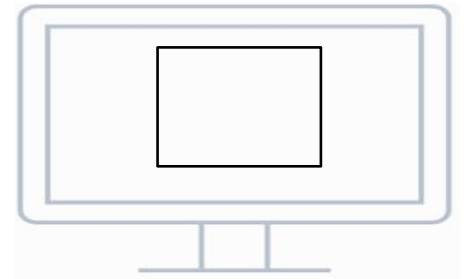
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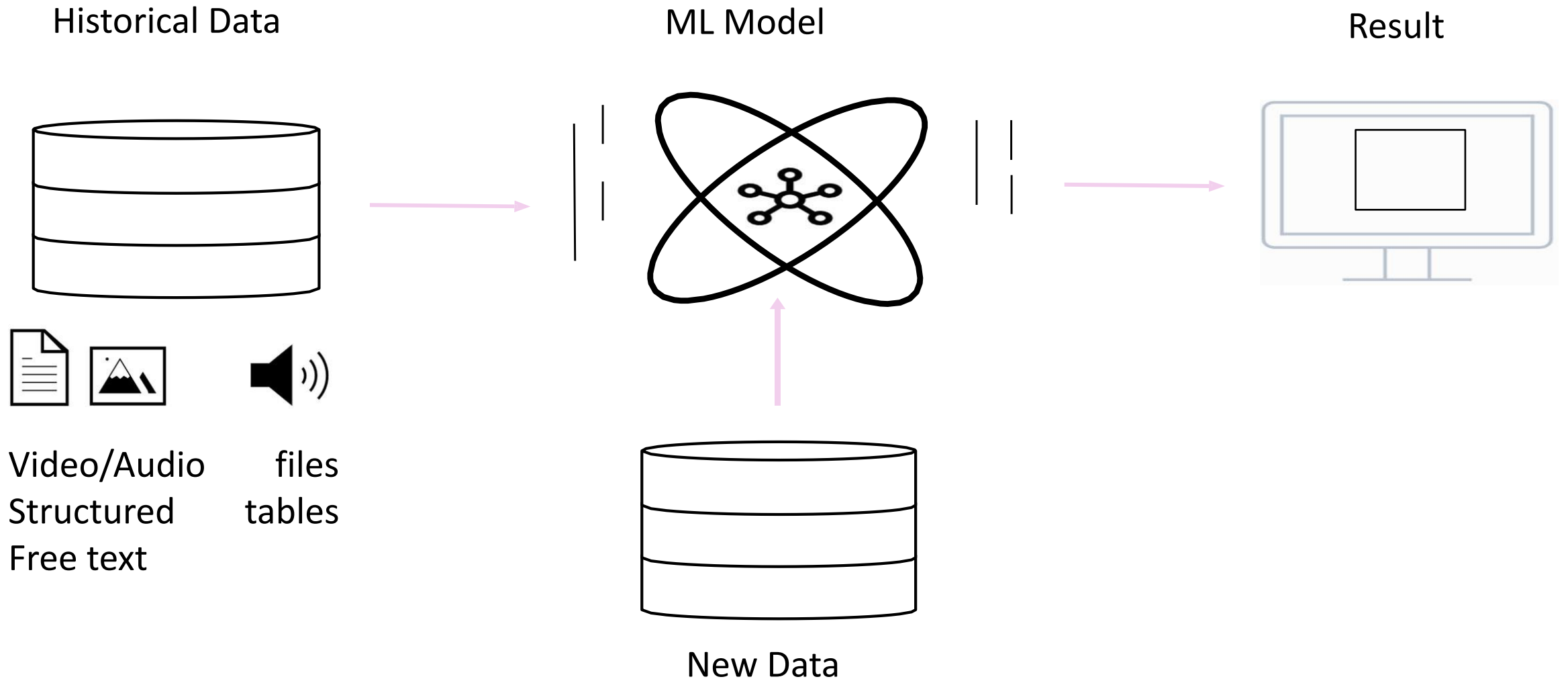
ML Model



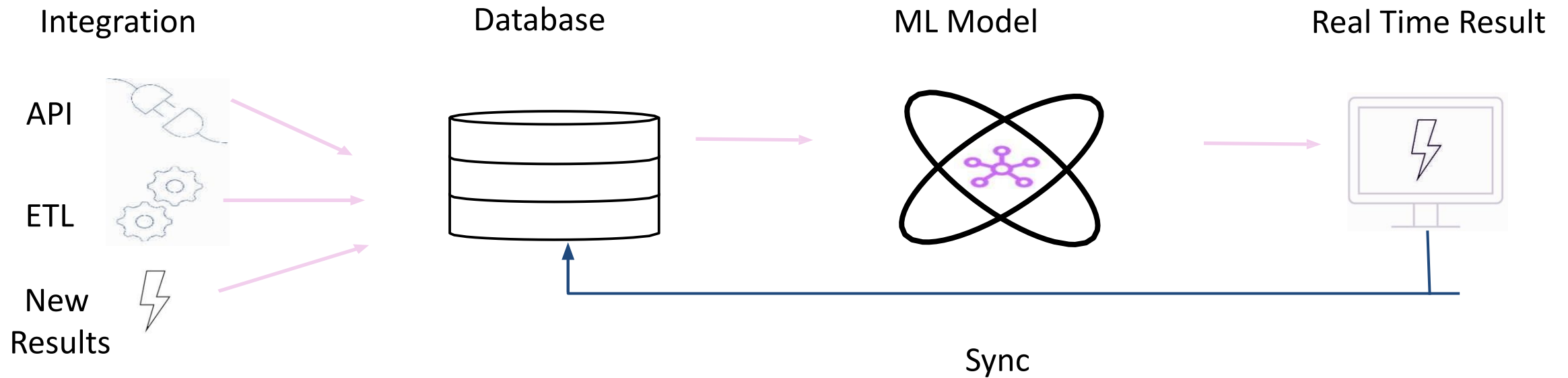
Result



# ML Architecture



# Production Pipeline



# Final Evaluation & Deployment

## **Deployment:**

Export the model as a serialized artifact (such as Pickle) and integrate it into a pipeline for real-time inference.

Track versions and ensure reproducibility.

# Monitoring

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## **Model Performance Tracking:**

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Continuously measuring Accuracy, Precision, or RMSE against real-world outcomes as they come in.

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## **Data & Concept Drift Detection:**

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- When the input data's distribution changes (e.g., a new sensor type starts sending different values).

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## **Data Drift Detection:**

- When the input data's distribution changes  
(e.g., a new sensor type starts sending different values).
- When the relationship between input and output changes  
(e.g., consumer shopping habits shift due to a new economic trend).

# Monitoring

## **Alerting Pipelines:**

Setting up automated triggers to notify engineers the moment performance drops below a threshold.

Tips

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  - b. If performance is not good in general- we need to go back to the EDA.

# Tips

1. Better data leads to better models! Data preprocessing is the most crucial part.
2. Prefer using pipelines to avoid data leakage
3. Check our errors.
  - a. Which records have underperforming results?
  - b. If performance is not good in general- we need to go back to the EDA.
4. Always pick the simplest model, and minimal preprocessing.

# Summary

1

### Understand Problem

Define the business problem and success metrics.

2

### Collect Data

Gather and prepare data for ML algorithms.

3

### Build Model

Choose and train ML algorithms.

4

### Evaluate Model

Test and refine the model's performance.

5

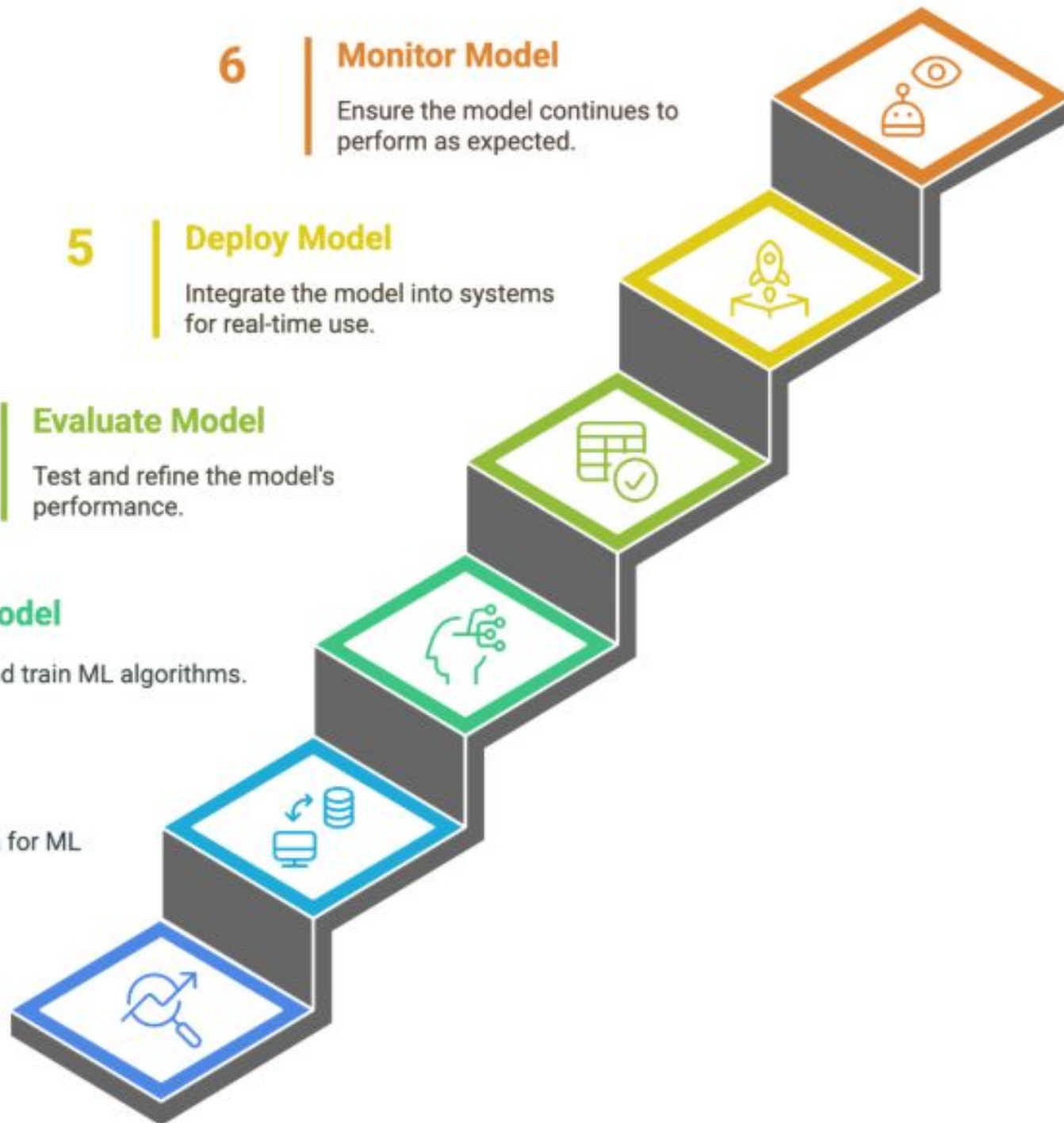
### Deploy Model

Integrate the model into systems for real-time use.

6

### Monitor Model

Ensure the model continues to perform as expected.



# Thank You!

Or Hirshman Shavit